Samuel Apoya

PhD Student in Computer Science @ UMBC | AI Infrastructure | Machine Learning | Software Engineering
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Education

University of Maryland, Baltimore County — Baltimore, MD
PhD in Computer Science — *June 2026 – Present*

Colby College — Waterville, ME
Bachelor of Arts in Computer Science — *Aug. 2022 – May 2026*

Technical Expertise

Programming: C++, Java, Python, Go, SQL, JavaScript, TypeScript
Machine Learning: PyTorch, TensorFlow, JAX, Transformers, Hugging Face, LoRA, CLIP, FAISS
Accelerated Computing: vLLM, Triton, GPU-accelerated training and inference
Systems & Infrastructure: Linux, Docker, Redis, AWS, REST APIs, GitHub Actions, Distributed Systems
Databases: PostgreSQL, MySQL, MongoDB

Experience

Software Engineering Intern — Oct. 2024 – May 2025
Davis Institute for Artificial Intelligence — *Waterville, ME*
- Engineered scalable Java backend services on AWS supporting concurrent NLP inference workloads, integrating MySQL-backed data services and production REST APIs.
- Developed a FAISS-based hybrid retrieval-augmented generation pipeline, improving answer accuracy by approximately 25%.
- Optimized backend throughput using Redis caching, SQL indexing, batching, and connection pooling, reducing p95 latency by 12%.
- Designed and deployed REST APIs with Docker-based GitHub Actions workflows, reducing deployment time by more than 50%.

Research Assistant – MLArtProject — May 2025 – May 2026
Insite Lab, Colby College — *Waterville, ME*
- Fine-tuned multimodal vision-language models using PyTorch and LoRA on Linux with GPU-accelerated training to generate spatially grounded image descriptions for blind and low-vision users.
- Developed backend infrastructure for a multimodal conversational AI system using Node.js, MongoDB, and React, implementing persistent chat sessions and REST APIs.
- Built CLIP-based evaluation pipelines and added structured logging to measure model performance and reliability across AI-assisted museum experiences.

Teaching Assistant — Feb. 2023 – Present
Colby College Computer Science Department — *Waterville, ME*
- Mentored students in Object-Oriented Design and Data Structures on design patterns, concurrency, algorithms, and debugging.
- Supported 200+ students across multiple semesters through weekly office hours, code reviews, and project troubleshooting.

Projects

FitFindr – Agentic AI Recommendation System | *Python, LLMs, Retrieval, Tool Calling* — 2026
- Built an autonomous AI agent that converts natural-language preferences and constraints into structured retrieval and recommendation workflows.
- Designed modular retrieval, reasoning, and explanation tools within an iterative agent loop, evaluated with automated tests.

CineRank | *FastAPI, PyTorch, Neural Ranking* — 2026
- Trained a PyTorch neural collaborative-filtering model for personalized movie recommendations and similar-item retrieval.
- Built a retrieval and ranking pipeline evaluated using Precision@K, Recall@K, and NDCG@K.

RAVE Neural Audio Model | *PyTorch, GPU-Accelerated Training* — 2025
- Trained a variational autoencoder for real-time neural audio synthesis using GPU-accelerated PyTorch and checkpoint-based experimentation.
- Evaluated reconstruction quality and exported a streaming-compatible model for an interactive research environment.

Leadership

President — Sep. 2024 – Present
Colby Hackers — *Waterville, ME*
- Organized ML and cloud-focused workshops, mentoring 10+ teams from idea to working prototype.
- Led planning and execution of a campus-wide hackathon with 60+ participants.